

# KARTHIK TATINENI

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## EDUCATION

### B.Tech in Electronics and Communication Engineering

2022 – 2026

Institute of Aeronautical Engineering, Hyderabad | CGPA: 7.2/10

## SKILLS

**Languages:** Python, JavaScript, SQL

**Web:** FastAPI, Node.js, REST APIs, React, HTML, CSS, Tailwind CSS

**AI/ML:** YOLO, CNN, RAG, MCP, AI Agents, Embedding

**DevOps/Cloud:** Docker, Kubernetes, CI/CD, AWS (EC2, S3, IAM), Azure (Container Apps, Container Registry), Azure DevOps, Azure Foundry, Cloudflare, Grafana

**Databases:** Firebase, Supabase, MongoDB, MySQL, PostgreSQL, Redis

**Tools:** Git, GitHub, JupyterLab, Anaconda, Postman, Vercel, Umami, Claude Code CLI, Copilot CLI, Hugging Face, Ollama, Nvidia NIM

## EXPERIENCE

### Freelance Software Developer

Jan 2025 – Present

- Built **KLVORA Fashion Hub**, a production-ready Next.js e-commerce platform with Firebase, Supabase, AI concierge, Three.js, secure authentication, checkout, inventory management, and order tracking. [\[Live\]](#)
- Built **Touchwood Interiors CMS**, a production-ready Next.js platform with Sanity CMS, Supabase, AI support chatbot, secure quotations, interactive 3D visualization, and architect portfolios. [\[Live\]](#)

## PROJECTS

### DisasterMind AI - Multi-Agent | *FastAPI, Next.js, NVIDIA NIM, PostgreSQL, pgvector* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Built a multi-agent disaster response platform that uses RAG and vector search over PostgreSQL/pgvector to deliver AI-powered emergency analysis and decision support.
- Integrated NVIDIA NIM-hosted models with a FastAPI backend and Next.js frontend, using text embeddings to retrieve relevant context for real-time agent reasoning during simulated disaster scenarios.

### Distributed College Event Management System | *React, Firebase, Docker, Nginx, Vercel, AWS* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Deployed a high-availability event platform handling 1,000+ live registrations with zero downtime across concurrent admin and participant sessions.
- Architected a scalable backend with microservices, Nginx load balancing, and Cloudflare CDN, achieving low response times under peak load.

### Real-Time Helmet & Triple-Ride Detection | *Python, YOLOv3, CNN, TensorFlow, PyTorch, OpenCV* | [\[GitHub\]](#) [\[Live Demo\]](#)

- Built a computer vision pipeline detecting helmet violations and triple-riding with 90%+ accuracy on test data; optimized to 25+ FPS for real-time traffic surveillance.
- Fine-tuned YOLOv3 on a custom-labeled dataset and implemented automated violation detection for law enforcement alert systems.

## PUBLICATIONS

T. Karthik Sai et al., **Helmet Violation and Triple-Ride Detection Using YOLOv3 and CNN**, ICMCER 2026 (Scopus Indexed) | [Certificate](#)

## CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals | [Certificate](#)
- MySQL & Database Management: Create, Manage & Query Databases | [Certificate](#)
- GenAI For Image & Video Creation | [Certificate](#)
- HTML & CSS: The Complete Web Development Guide | [Certificate](#)
- Microsoft AI Skills Fest 2026 | [Credly Badge](#)

## EXTRA-CURRICULAR

- Hosted and organised the official **IARE Consortium** event for 1,000+ participants; designed and developed the end-to-end registration platform.
- Core Member and Logistics Head**, NSS – led community logistics for college and district level programs.